AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

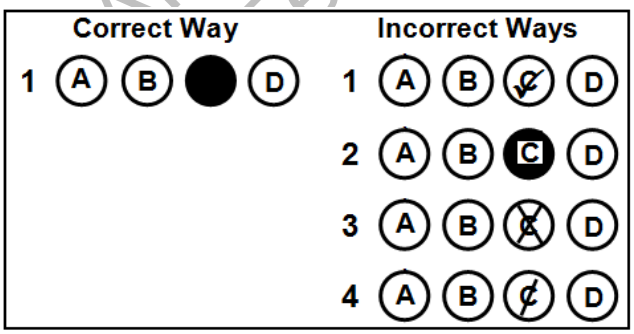
Business Statistics Paper I

Time: 55 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols have their usual meanings.

Measures of Central Tendency and Quartiles

$$\overline{X} = \frac{\sum wx}{\sum w} \qquad \overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{\sum f} \qquad \mathrm{Q}_i = l + \frac{h}{f}\left(\frac{n \times i}{4} - c\right)$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Measures of Dispersion

$$\text{variance} = \frac{\sum \left(x - \overline{X}\right)^2}{n} \qquad \text{SD} = \sqrt{\frac{\sum \left(x - \overline{X}\right)^2}{n}}$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{variance} = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2 \qquad SD = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

$$\text{variance} = \frac{\sum f\left(x - \bar{x}\right)^2}{\sum f} \qquad SD = \sqrt{\frac{\sum f\left(x - \bar{x}\right)^2}{\sum f}}$$

$$\mathrm{R} = x_{\max} - x_{\min} \qquad \text{Coefficient of Variance} = \frac{SD}{\overline{X}} \times 100$$

Index Number

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100 \qquad P_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100 \qquad P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times 100$$

Counting Techniques and Probability

$${}^{n}P_r = \frac{n!}{(n-r)!} \qquad {}^{n}C_r = \frac{n!}{(n-r)!r!}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(A) \times P(B) \qquad P(A \cap B) = P(B) \times P(A \mid B)$$

Scatter Diagram and Correlation

$$r = \frac{n \sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{\left[n \sum X^2 - \left(\sum X\right)^2\right]\left[n \sum Y^2 - \left(\sum Y\right)^2\right]}}$$

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1. A school wants to estimate the average math score of a class of 50 students for a 75 marks test. In a random sample of 25 students, the sample average score is 40 marks.

   The total number of students in a class represents

   A. sample.
   B. statistic.
   C. parametre.
   D. population.

2. A 5th grade classroom teacher wants to know how his students interact in a classroom setting.

   The most appropriate method of collecting data for the given requirement will be

   A. individual interviews of parents.
   B. discussion with other teachers.
   C. previous academic reports.
   D. classroom observations.

3. State which of the following represent(s) discrete data?

   I. Speed of an automobile in miles per second
   II. Amount (Rs) of electricity bill of a consumer
   III. Measurement of rainfall (*mm*) in a city during various months of the year

   A. I only
   B. I and II
   C. III only
   D. II and III

Use the given information to answer Q.4, Q.5 and Q.6.

The given frequency table shows the low-density lipoprotein (LDL) cholesterol of 30 persons aged between 25 to 30 years.

LDL Range *mg/dL*	Number of Persons
60 – 69	2
70 – 79	6
80 – 89	9
90 – 99	7
100 – 109	6

4. The number of persons whose LDL cholesterol is less than or equal to 89 *mg/dL* is

   A. 8
   B. 9
   C. 10
   D. 17

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5. The upper class boundary of third class interval will be

   A. 79
   B. 79.5
   C. 89
   D. 89.5

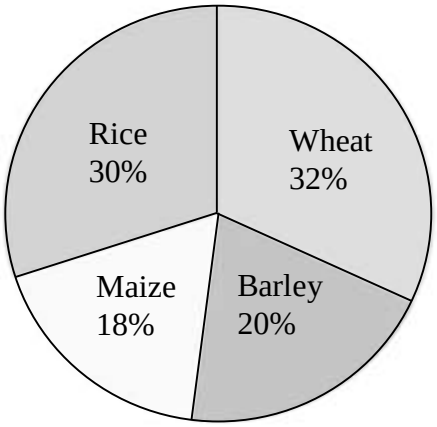
6. The relative frequency for the second class will be

   A. 0.20
   B. 0.26
   C. 0.73
   D. 0.80

7. The graphical representation of continuous data can only be done by using

   A. pie chart only.
   B. histogram only.
   C. bar chart and histogram.
   D. simple bar chart and pie chart.

8. A shopkeeper has four commodities, maize, rice, wheat and barley as represented in the given pie chart. Each commodity is measured in kilogram.



   What is the angle associated with wheat in the given pie chart, which is 32% of the total commodities?

   A. 4.44^o
   B. 8.89^o
   C. 57.6^o
   D. 115.2^o

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9. If the weighted mean of 8 observations in a grouped data is 40 and $\sum wx = 1{,}505$, then $\sum w$ is equal to

   A. 4.70
   B. 37.62
   C. 188.12
   D. 60,200

10. The average driving speed of 8 vehicles on a highway is *a km/ h*. If due to rain, every vehicle slows down by *b km/ h*, then the new average driving speed will be

   A. $a - b$ *km/ h.*
   B. $a + b$ *km/ h.*
   C. $\frac{a-b}{8}$ *km/ h.*
   D. $\frac{a+b}{8}$ *km/ h.*

Use the given information to answer Q.11 and Q.12.

The data set 2, 5, 7, x, $2x$, $2x + 2$, 25, 50 is arranged in ascending order.

11. If the median of the data set is 13.5, then the value of x will be

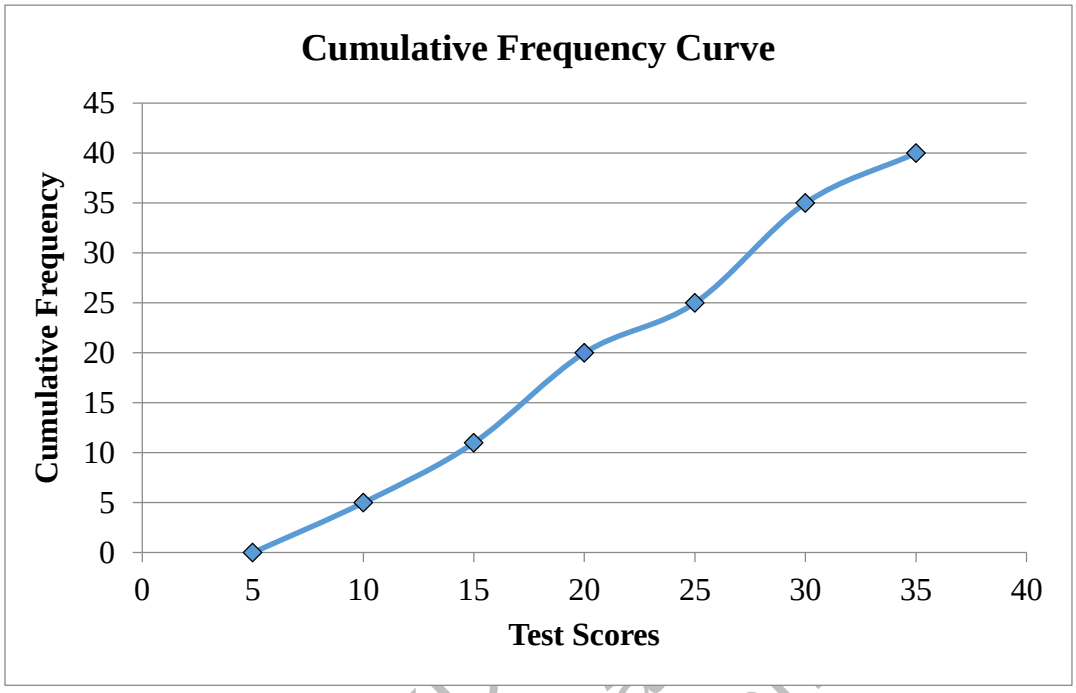
   A. 7
   B. 9
   C. 19
   D. 27

12. Assuming that the value of x is equal to 8, then the value of 3rd Quartile would be

   A. 8
   B. 12
   C. 14
   D. 18

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13. The given graph shows a cumulative frequency curve of an aptitude test scores.

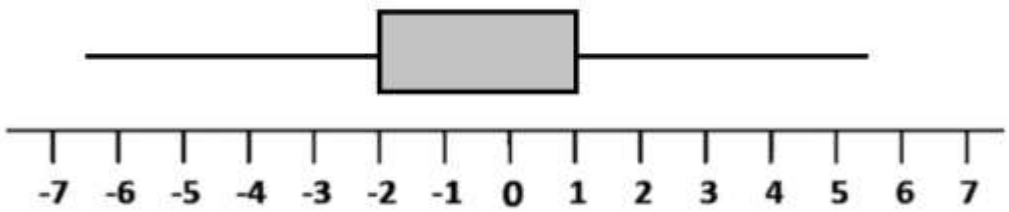


The number of students who scored between 20 to 35 is

A. 15
B. 20
C. 30
D. 80

Use the given diagram to answer Q.14 and Q.15.

The given diagram shows the box and whisker diagram (5 number summary) for a certain data



14. The interquartile range is

A. -3
B. -0.5
C. 1.5
D. 3

15. The lower extreme value is

A. -6.5
B. -2
C. 2.5
D. 5.5

16. If for a set of data, $\sum\left(x-\overline{X}\right)^2 = 250$ and $\sigma = 5$, then the value of *n* is equal to

A. 10
B. 50
C. $\sqrt{50}$
D. $\sqrt{10}$

17. The standard deviation of the data set a, a, a, a, a, and a is

A. 0
B. a
C. $\sqrt{a}$
D. $\pm\sqrt{a}$

18. If variance $\left(\frac{X}{0.02}\right)$ = 0.4, then $\frac{1}{(0.02)^2}$ (variance *X*) is equal to

A. 1000
B. 50
C. 0.4
D. 0.16

19. If $\sum X = \sum X^2 = 4$ and $\bar{x}$ = 1 for four observations, then the coefficient of variance will be

A. 400
B. 200
C. 25
D. 0

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20. The given table shows the average price (in Rupees per kilograms) of the most used vegetables in Karachi for the years 2000 and 2015.

Vegetable	Average Price (Rs/ kg)	
	2000	2015
Onion	15	22
Potatoes	12	20
Tomatoes	30	35

(**Note**: Based on fictitious data)

The aggregate price index (of these commodities) for the year 2015, taking year 2000 as base period, is

A. 174.1%.
B. 135.1%.
C. 74.1%.
D. 35.1%.

21. If the simple price index of a certain commodity for the year 2015, taking year 1995 as base period, is 171.4%, then it means that the price of this commodity in 2015, as compared to 1995, is

A. 71.4% higher.
B. 17.14% higher.
C. 71.4% lower.
D. 17.14% lower.

22. How many four digits can be formed without replacement from the 6 digits 1, 2, 3, 4, 5, 6?

A. ${}^6P_4$
B. ${}^6P_3$
C. ${}^5P_4$
D. ${}^5P_3$

23. The number of possible arrangements of the word STATISTICS is

A. $\frac{10!}{3!2!1!}$.
B. $\frac{10!}{9!2!1!}$.
C. $\frac{10!}{3!3!1!2!1!}$.
D. $\frac{10!}{3!6!1!}$.

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24. A carton of 11 rechargeable batteries contains two that are defective. An inspector randomly selects 4 of the batteries. The number of selections that he can do such that none of the selected batteries is defected will be

A. 330
B. 126
C. 55
D. 36

25. There are 50 people in a que. Ali is the 22nd person and Sara is the 17th person. If a person is chosen at random, the probability that the person is standing between Ali and Sara is

A. 0.08
B. 0.1
C. 0.12
D. 0.25

26. If A and B are mutually exclusive events where $P(A) = 0.45$ and $P(B') = 0.45$, then $P(A \text{ or } B)$ is

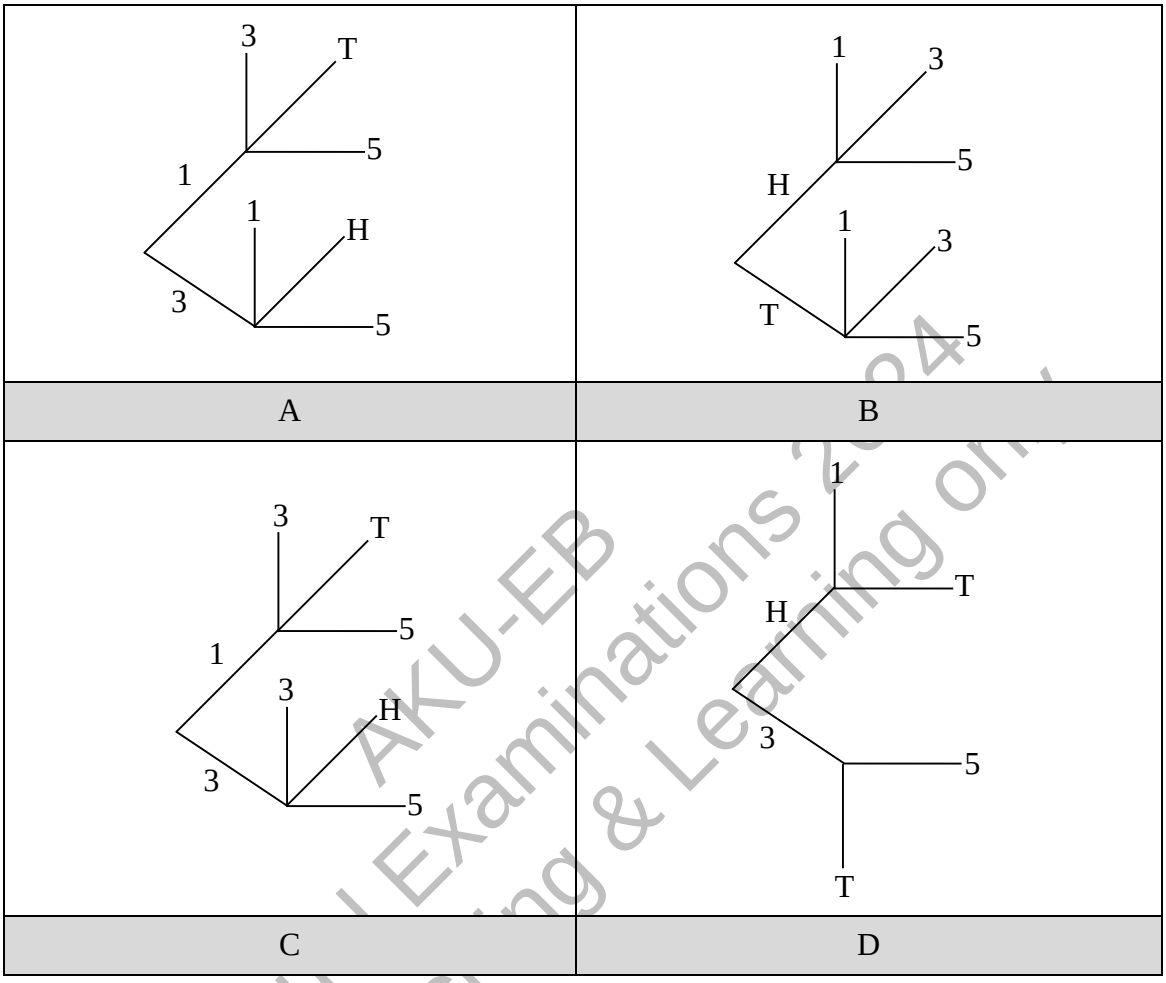
A. 1.10
B. 1.00
C. 0.55
D. 0.45

27. The events M and N are independent. If $P(M/N) = \frac{1-x}{5}$, $P(N) = \frac{2}{5}$ and $P(M \cap N) = \frac{6}{25}$, then the value of x is

A. -2
B. -4
C. 2
D. 4

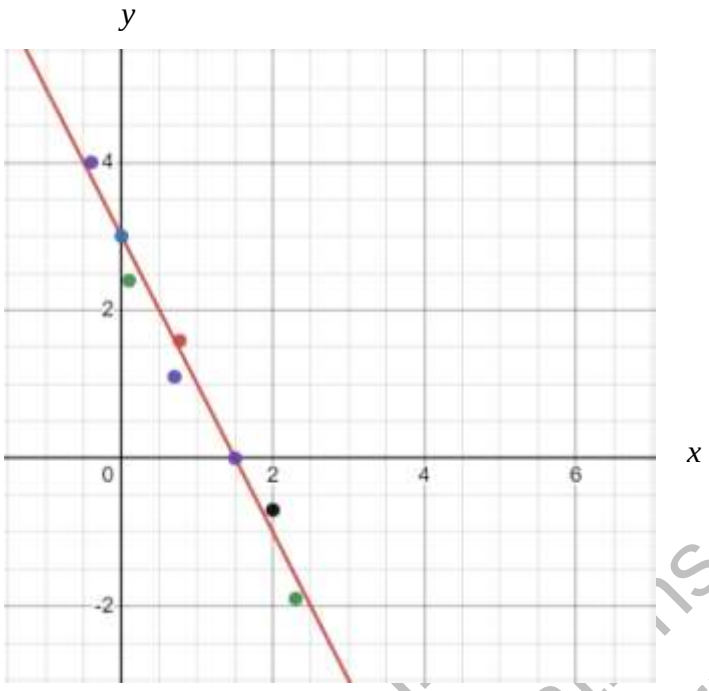
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28. A spinner having numbers 1, 3, 5 and a coin are thrown together. The tree diagram which illustrates the possible outcomes of these two will be



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29. The given scatter diagram shows the location of points (x, y).



The coefficient of correlation will be

A. strong positive.
B. strong negative.
C. perfect positive.
D. perfect negative.

30. If there is a perfect inverse correlation between two variables, then the value of coefficient of correlation r will be

A. -1
B. -0.5
C. 0.5
D. 1

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END OF PAPER

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